ML-SemReg: Boosting Point Cloud Registration with Multi-level Semantic Consistency

Shaocheng Yan[1], Pengcheng Shi[2], and Jiayuan Li[1,†]

[1] School of Remote Sensing and Information Engineering, Wuhan University
[2] School of Computer Science, Wuhan University
{shaochengyan, shipc_2021, ljy_whu_2012}@whu.edu.cn

Abstract. Recent advances in point cloud registration mostly leverage geometric information. Although these methods have yielded promising results, they still struggle with problems of low overlap, thus limiting their practical usage. In this paper, we propose ML-SemReg, a plug-and-play point cloud registration framework that fully exploits semantic information. Our key insight is that mismatches can be categorized into two types, i.e., inter- and intra-class, after rendering semantic clues, and can be well addressed by utilizing multi-level semantic consistency. We first propose a Group Matching module to address inter-class mismatching, outputting multiple matching groups that inherently satisfy Local Semantic Consistency. For each group, a Mask Matching module based on Scene Semantic Consistency is then introduced to suppress intra-class mismatching. Benefit from those two modules, ML-SemReg generates correspondences with a high inlier ratio. Extensive experiments demonstrate excellent performance and robustness of ML-SemReg, e.g., in hard-cases of the KITTI dataset, the Registration Recall of MAC increases by almost 34 percentage points when our ML-SemReg is equipped. Code is available at `https://github.com/Laka-3DV/ML-SemReg`

Keywords: Point cloud registration · Semantic information · 3D matching

1 Introduction

Point cloud registration (PCR), a technique that aligns two point clouds from different scans of the same scene, is crucial for applications such as 3D reconstruction [10], simultaneous localization and mapping (SLAM) [16], and augmented reality [4]. Correspondence-based registration does not rely on initialization and has received more attention recently. It generally consists of two major stages: keypoint matching and transformation estimation [9, 31, 33, 42, 45, 53, 57]. The latter relies on correspondences outputted by the former. Even though the estimator has achieved good robustness, registration fails when facing small number of inliers or a low inlier ratio [9,57]. Thus, obtaining high-quality correspondences is an urgent problem to be solved.

† indicates the corresponding author.

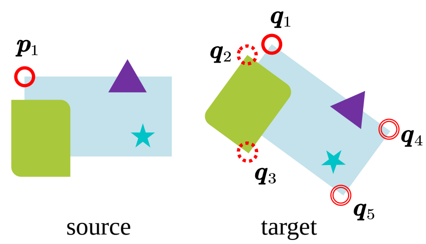


Fig. 1: Inter-class and intra-class mismatching. We use various shapes (such as rectangles, triangles, etc.) to represent different object categories (semantic labels) and depict them in different colors. Keypoint (inside the red circle) $\boldsymbol{p}_1$ shares the same local geometric features with target keypoints $\boldsymbol{q}_{1\sim5}$, resulting in inter-class mismatching between $\boldsymbol{p}_1$ and $\boldsymbol{q}_{2\sim3}$, and intra-class mismatching between $\boldsymbol{p}_1$ and $\boldsymbol{q}_{4\sim5}$.

Different from image keypoint matching, 3D matching becomes extremely difficult due to problems such as lack of texture, uneven density, and disordered organization [42, 44]. Therefore, a substantial body of research has concentrated on developing more robust local descriptors, aiming to improve the inlier ratio of correspondences. [2, 25, 31, 42, 53]. Despite these methods have achieved excellent performance with the development of deep learning, the ubiquitous local geometric similarities in point clouds still pose great challenges to them [31]. For example, as shown in Fig. 1, keypoint $\boldsymbol{p}_1$ shares identical right-angle local features with candidates $\boldsymbol{q}_{1\sim5}$, resulting in a correct matching probability of just $\frac{1}{5}$, and the probability of all right angles being perfectly matched is even lower, as low as $\frac{1}{5!} \approx 0.83\%$.

Semantic information can help gain insights into mismatches. As illustrated in Fig. 1, mismatches after rendering semantic information can be categorized into two types: (1) Inter-class mismatching: Even though two keypoints are similar in geometric information, they are located in different semantic categories, e.g., $\boldsymbol{p}_1$ and $\boldsymbol{q}_{2\sim3}$. (2) Intra-class mismatching: Although keypoint pairs are both similar in geometric information and belong to the same semantic category, they are in different scene contexts, such as $\boldsymbol{p}_1$ and $\boldsymbol{q}_{4\sim5}$. Recently, semantic-assisted 3D matching has gradually received attention [27, 32, 51, 58]. These methods only match keypoints with the same label, thus reducing inter-class mismatching. Unfortunately, some inliers may have different semantic labels due to unreliable semantic segmentation. Therefore, these methods may lose a significant number of inliers, resulting in registration failure. Furthermore, they cannot address intra-class mismatching as they ignore the scene information of keypoints.

To address the aforementioned challenges, we introduce a PCR framework named ML-SemReg. Our key insight is that inter- and intra-class mismatches can be effectively addressed by leveraging multi-level semantic consistency. For local-level, we start by introducing a robust Local Semantic Consistency (LS-Consistency), which fully exploits local semantic relations between keypoints. Next, to tackle inter-class mismatching, we propose a Group Matching (GM) module to categorize keypoints into matching groups that inherently satisfy LS-

Consistency. For scene-level, we introduce a descriptor called Binary Multi-Ring Semantic Signature (BMR-SS) to capture scene semantic information. Based on the BMR-SS, a Mask Matching (MM) module is introduced for each group to maintain Scene Semantic Consistency (SS-Consistency), thereby suppressing intra-class mismatching. Finally, the ML-SemReg framework integrates GM and MM modules, thus ensuring multi-level semantic consistency. In summary, the contributions of our work are three-fold:

- A Group Matching module is introduced to address inter-class mismatching, leveraging our robust LS-Consistency.
- A Mask Matching module is proposed to address intra-class mismatching based on SS-Consistency by utilizing our scene-aware descriptor BMR-SS.
- A plug-and-play PCR framework named ML-SemReg is presented to integrate Group Matching and Mask Matching modules, which ensures multi-level semantic consistency to generate high-quality correspondences.

2 Related Work

3D Keypoint Matching. Different from ICP-type methods [5, 6, 34, 37] that establish correspondences by nearest searching in coordinate space, keypoint matching uses descriptor distances in feature space. Traditional approaches [1, 7, 18, 20, 21, 24, 35, 36, 40, 55] rely on manually designed local patch geometric descriptors. With the advancement of deep learning, learned 3D feature descriptors have outperformed hand-crafted ones. A pioneering work is 3DMatch [56], which uses a siamese convolutional network for description. Recently, many networks have attempted to improve the performance by resorting to rotation invariant modules [2,14,15,42,49], fully convolution modules [12,25,31], and transformer modules [33,50,53], or by designing detect-description [44] and encoding-decoding frameworks [23,33,52]. Although these methods have achieved remarkable performance improvements, they still face numerous challenges caused by local geometric similarities.

Robust Model Estimation. Traditional robust estimation methods, such as RANSAC [17], tend to converge slowly and exhibit instability when dealing with a high proportion of outliers. To address this issue, several algorithms [47, 48, 51, 57] employ length consistency to construct graphs for searching for a maximal clique of inliers and subsequently use robust estimators for transformation estimation. Deep estimators [11, 30, 45] typically consist of an inlier/outlier classification network and a transformation estimation network, which have shown improvements in both accuracy and speed. While these methods excel at handling low inlier ratios, they may encounter challenges when the number of initial correspondences is extremely small. In such cases, there are too many outliers to vote for inliers [48, 57], or the maximum clique assumption might not hold [32], rendering these algorithms ineffective. In contrast, our approach effectively retains inliers and achieves a high inlier ratio, thereby enhancing the robustness and performance of these methods.

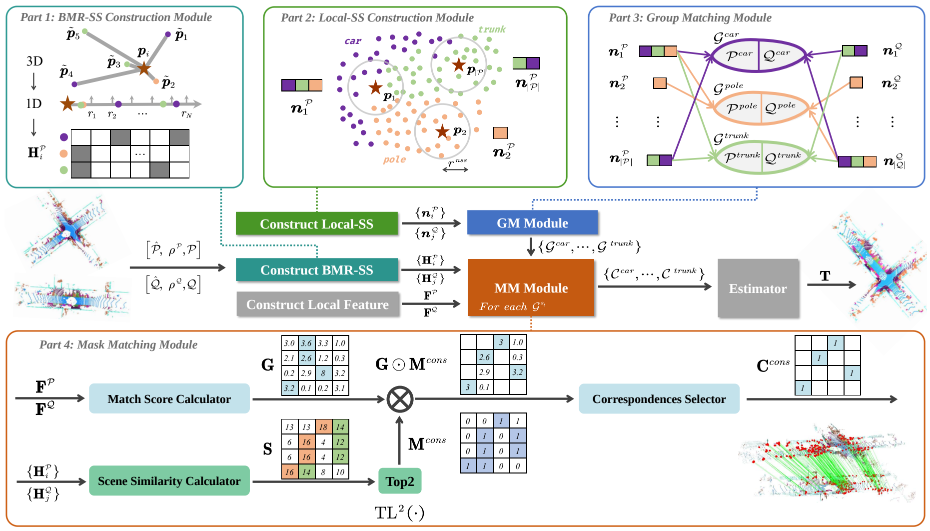


Fig. 2: ML-SemReg Framework. For keypoints from the source and target point clouds, the proposed ML-SemReg (1) constructs the BMR-SS and Local-SS of each keypoint to perceive scene and local semantic information, respectively. (2) Simultaneously, the local geometry feature is calculated by a descriptor (e.g., FPFH [35]). (3) For each semantic category, the GM module produces a matching group, in which the keypoints satisfy LS-Consistency among each other. (4) For each matching group, the MM module constructs a scene consistency mask, outputting sub-correspondences sets with multi-level semantic consistency. (4) Finally, the union of subsets produces high-quality correspondences, which are used to estimate the rigid transformation matrix.

Semantic-assisted Matching. Existing solutions can be categorized into keypoint-level and entity-level methods. Keypoint-level methods directly utilize the semantic labels of keypoints to avoid matching between different categories [3, 8, 27, 41, 54]. For instance, SARNet [27] employs a semantic mask to filter out correspondences with different categories. In contrast, entity-level algorithms first construct semantic clusters (or entities) and then perform matching on them [28,32,51,58]. They fully exploit the saliency of entities in the scene, thus being more robust to semantic noise. Segregator [3] combines both entity-level and keypoint-level approaches, incorporating the advantages of both. However, these methods ignore the scene information of keypoints, making them very sensitive to intra-class mismatching. In contrast, our method explores Scene Semantic Consistency, thereby effectively addressing intra-class mismatching problem.

3 Method

3.1 Problem Formulation

Given two point clouds $\hat{\mathcal{P}} \subset \mathbb{R}^3$ (source) and $\hat{\mathcal{Q}} \subset \mathbb{R}^3$ (target), our goal is to find a sufficient number of inliers to align them. The framework of the ML-

SemReg is shown in Fig. 2, which takes $\hat{\mathcal{P}}$, $\hat{\mathcal{Q}}$, $\rho^{\mathcal{P}}$, $\rho^{\mathcal{Q}}$, $\mathcal{P}$, and $\mathcal{Q}$ as its inputs, where $\mathcal{P} = \{\boldsymbol{p}_i\}$ and $\mathcal{Q} = \{\boldsymbol{q}_j\}$ are keypoints randomly selected from $\hat{\mathcal{P}}$ and $\hat{\mathcal{Q}}$, respectively. $\rho^{\mathcal{T}} : \mathcal{T} \to \mathcal{S}, \mathcal{T} \in \{\mathcal{P}, \mathcal{Q}\}$ represents label mapping function (e.g., PTv2 [43]), where $\mathcal{S} = \{s_t\}$ is a set of semantic labels. Our method comprises the Group Matching module (Sec. 3.2) and the Mask Matching module (Sec. 3.3), which address inter-class and intra-class mismatching, respectively.

3.2 Group Matching

To solve inter-class mismatching, previous works [3, 8, 22, 27] strictly avoid keypoint matching between two categories. However, this approach is not robust when faced with semantic noise, leading to a poor number of inliers and ratio. We address this problem by relaxing the matching of keypoints *within the same category* to a more robust *within matching groups*.

Local Semantic Consistency. We first propose LS-Consistency as follows:

$$\boldsymbol{n}_i^{\mathcal{P}} \cap \boldsymbol{n}_j^{\mathcal{Q}} \neq \varnothing \to \mathit{\Gamma}^{LS}\left(\boldsymbol{p}_i, \boldsymbol{q}_j\right) = 1. \tag{1}$$

where $\boldsymbol{n}_i^{\mathcal{P}} = \left\{\rho^{\mathcal{P}}\left(\boldsymbol{p}_j\right) \| \|\boldsymbol{p}_j - \boldsymbol{p}_i\|_2 \leq r^{local}, \boldsymbol{p}_j \in \mathcal{P}\right\}$ denotes the Local Semantic Signature (Local-SS) under search radius r^{local}. The indicator function $\mathit{\Gamma}^{LS}$: $\mathcal{P} \times \mathcal{Q} \to \{0, 1\}$ signifies whether points $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$ satisfy the LS-Consistency.

Group Matching. It is easy to prove that when the search radius r^{local} is large enough (e.g., 2 $\times$ the inlier threshold), a correspondence can be classified as an inter-class mismatching (outlier) if two keypoints cannot satisfy the LS-Consistency. Thus, we can solve inter-class mismatching by avoiding the matching of keypoints that do not satisfy LS-Consistency. To implement this, we first define a Matching Group as a combination of two keypoint sets:

$$\mathcal{G}^{s_t} = (\mathcal{P}^{s_t}, \mathcal{Q}^{s_t}), \tag{2}$$

where $\mathcal{P}^{s_t} = \left\{\boldsymbol{p}_i | s_t \in \boldsymbol{n}_i^{\mathcal{P}},\ \boldsymbol{p}_i \in \mathcal{P}\right\}$ is a sub keypoint set of source keypoints of label s_t, and $\mathcal{Q}^{s_t}$ is a sub keypoint set of target keypoints. Obviously, keypoints in $\mathcal{G}^{s_t}$ satisfy LS-Consistency because each of their Local-SS contains s_t. Conversely, if two keypoints do not exist in any matching group, they cannot be matched because they do not satisfy LS-Consistency. Therefore, by just matching keypoints in each Matching Group and then merging them, a final set of correspondences without inter-class mismatching can be obtained:

$$\mathcal{C} = \bigcup_{s_t \in \mathcal{S}} \mathrm{MF}\left(\mathcal{G}^{s_t}\right), \tag{3}$$

where $\mathrm{MF}(\cdot)$ represents a matching function like the nearest neighbor matcher.

3.3 Mask Matching

In this section, we first construct the Binary Multi-ring Semantic Signature (BMR-SS) to quantify scene similarity. Based on scene similarity, we propose the

Mask Matching (MM) module to mitigate intra-class mismatching by selectively discarding correspondences involving keypoint pairs with low scene similarity.

Construction of BMR-SS. Inspired by [2, 26], which construct descriptors through spatial division, we adopt this strategy to construct BMR-SS for encoding scene information. We first construct $Z^{\mathcal{P}}$ clusters of semantic instances (car, traffic sign, pole, etc.) using euclidean clustering with different radii [51]. Then, the landmark set is defined as the collection of cluster centers:

$$\mathcal{L}^{\mathcal{P}} = \{\tilde{\boldsymbol{p}}_i | i = 1 \cdots Z^{\mathcal{P}}, \tilde{\boldsymbol{p}}_i \in \mathbb{R}^3\}, \tag{4}$$

where $\tilde{\boldsymbol{p}}_i$ is centroid of i-th cluster. Next, we divide the scene into N rings of width L centered on keypoint, where the radius range of the i-th ring is $[r_{i-1}, r_i], r_i = iL, \forall i = 1 \cdots N$. Then, the BMR-SS $\mathbf{H}_i^{\mathcal{P}} \in \{0,1\}^{|\mathcal{S}| \times N}$ is filled by:

$$[\mathbf{H}_i^{\mathcal{P}}]_{tk} = \begin{cases} 1 & \text{if } s_t \in \varUpsilon_k\left(\boldsymbol{p}_i\right) \\ 0 & \text{otherwise} \end{cases}, \quad t = 1, \cdots, |\mathcal{S}|, \quad k = 1, \cdots, N, \tag{5}$$

where $\varUpsilon_k\left(\boldsymbol{p}_i\right) = \{\rho^{\mathcal{P}}\left(\tilde{\boldsymbol{p}}\right) | r_{k-1} \leq \|\tilde{\boldsymbol{p}} - \boldsymbol{p}_i\|_2 < r_k, \tilde{\boldsymbol{p}} \in \mathcal{L}^{\mathcal{P}}\}$ query the label set of landmark within the k-th ring around $\boldsymbol{p}_i$. The interpretation of Eq. (5) is that if a landmark with category s_t is in the k-th ring of $\boldsymbol{p}_i$, then $[\mathbf{H}_i^{\mathcal{P}}]_{tk} = 1$. Note that the encoding $\mathbf{H}_i^{\mathcal{P}}$ relies solely on distance (1D), thus inherently possessing rotation invariance required for descriptors [14, 31, 56].

Scene Semantic Similarity. Semantic instances in a scene can provide clues for scene similarity. For instance, if two keypoints are both approximately 2 meters away from a traffic light, they may be in similar positions. Building upon this observation, we propose to measure the scene semantic similarity of two keypoints by counting occurrences of the same landmark label within the same distance range around them. Moreover, if there exists a semantic label s_t in the k-th ring for both $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$, then $[\mathbf{H}_i^{\mathcal{P}}]_{tk} = [\mathbf{H}_j^{\mathcal{Q}}]_{tk} = 1$. Therefore, the scene semantic similarity between $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$ can be expressed as:

$$\varTheta\left(\boldsymbol{p}_i, \boldsymbol{q}_j\right) = \sum_{t=1}^{|\mathcal{S}|} \sum_{k=1}^{N} \left[\mathbf{H}_i^{\mathcal{P}}\right]_{tk} \times \left[\mathbf{H}_j^{\mathcal{Q}}\right]_{tk} = |\mathbf{H}_i^{\mathcal{P}} \odot \mathbf{H}_j^{\mathcal{Q}}|. \tag{6}$$

Fig. 3 illustrates the calculation of scene semantic similarity between $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$ (the same location in different LiDAR scans) using BMR-SS.

Mask Matching. The MM module aims to integrate scene semantic similarity into local feature matching, addressing intra-class mismatching. To combine the two levels of features, we initially formalize feature matching methods as a selection process, utilizing a match score matrix as input:

$$\mathbf{C} = \theta(\mathbf{G}), \tag{7}$$

where $\mathbf{G} \in [0,1]^{|\mathcal{P}| \times |\mathcal{Q}|}$ denotes the match score matrix based on local descriptor similarities, such as the cosine similarity of the GEDI descriptor. $\theta : [0,1]^{|\mathcal{P}| \times |\mathcal{Q}|} \rightarrow \{0,1\}^{|\mathcal{P}| \times |\mathcal{Q}|}$ is binary function, which selects matches with

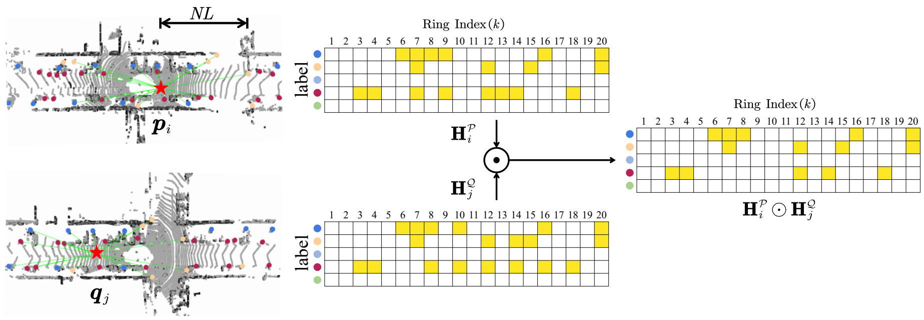


Fig. 3: Same locations exhibit scene similarity across different LiDAR scans. Keypoint $\boldsymbol{p}_i$ is connected to the landmarks (centroid of object) within its maximum receptive field radius NL. For $\mathbf{H}_i^{\mathcal{P}}$, the t-th row and k-column correspond to semantic label s_t and k-th ring, respectively. The final scene similarity between $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$ is 14 as $\Theta\left(\boldsymbol{p}_i, \boldsymbol{q}_j\right) = \left|\mathbf{H}_i^{\mathcal{P}} \odot \mathbf{H}_j^{\mathcal{Q}}\right| = 14$.

high match scores, outputting $\mathbf{C} \in \{0,1\}^{|\mathcal{P}|\times|\mathcal{Q}|}$, where $\mathbf{C}_{ij} = 1$ means $(\boldsymbol{p}_i, \boldsymbol{q}_j)$ is a putative inlier. Subsequently, by reweighting the match score matrix using an elaborate scene consistency mask $\mathbf{M}^{cons} \in \{0,1\}^{|\mathcal{P}|\times|\mathcal{Q}|}$, we can effectively suppress intra-class mismatching. To construct $\mathbf{M}^{cons}$, We first calculate the scene similarity matrix $\mathbf{S} \in \mathbb{N}^{|\mathcal{P}|\times|\mathcal{Q}|}$, where $[\mathbf{S}]_{ij} = \Theta(\boldsymbol{p}_i, \boldsymbol{q}_j)$ represents the scene similarity between keypoints $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$. Next, a binary selection function $\mathrm{TL}^K(\cdot)$ is used to set the highest K values in each row of $\mathbf{S}$ to 1, while the remaining values are set to 0. This process yields the scene consistency mask $\mathbf{M}^{cons}$, where $[\mathbf{M}^{cons}]_{ij} = 1$ means that $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$ satisfy Scene Semantic Consistency (SS-Consistency). Finally, based on the masked match score matrix $\mathbf{G} \odot \mathbf{M}^{cons}$, correspondence selection can be represented as an extension of Eq. (7):

$$\mathbf{C}^{cons} = \theta\left(\mathbf{G} \odot \mathbf{M}^{cons}\right). \tag{8}$$

By applying Mask Matching to each group generated by the Group Matching module, we can produce a high-quality set of correspondences with both local and Scene Semantic Consistency:

$$\mathcal{C} = \bigcup_{\mathcal{G}^{s_t} \in \mathrm{GM}(\mathcal{P},\mathcal{Q})} \mathrm{MM}\left(\mathcal{G}^{s_t}\right). \tag{9}$$

4 Experiment

4.1 Datasets and Experimental Setup

Outdoor Benchmark. The KITTI-Odometry dataset [19] is used for outdoor evaluation. Following [12, 45, 57], we construct an easy-type dataset using point cloud pairs with a 10m translation offset. Additionally, we construct medium-type and hard-type datasets with translational distances of 20m and 30m, respectively, which can comprehensively assess registration performance in scenarios with low overlap ratios. Note that we conduct the evaluation on all KITTI sequences instead of only sequences 8∼10 [9, 12, 33, 57]. Hence, we obtain 2096, 1060, and 702 point cloud pairs for registration in these three types of datasets.

Indoor Benchmark. The ScanNet dataset [13] is used for indoor evaluation. Since the raw ScanNet dataset is huge, we apply a selection criterion to retain pairs whose overlap ratios falling within a specific range in the sub-dataset *scannet_frames_25k*, resulting in 1142 point cloud pairs with overlap ratios ranging from 50% to 60% and 980 pairs with overlap ratios ranging from 40% to 50%.

Implementation Details. Our algorithm is implemented in Python using the Open3D library [59]. We use PTv2 [43] to generate semantic labels. In the Group Matching module, r^{local} is set to $0.8m$ for KITTI and $5cm$ for ScanNet. During the construction of BMR-SS, the number of ring (N) and the length of ring (L) are set to 33 and $1.5m$ for KITTI, while 10 and $20cm$ for ScanNet. In the MM module, K is set to 2 and 3 for KITTI and ScanNet, respectively. All experiments are conducted on a PC equipped with an Intel(R) i7-13700KF processor, 32GB of RAM, and a GeForce RTX 4090 GPU.

Evaluation Criteria. Following [12], we evaluate the performance of correspondence set using Inlier Ratio (IR) [31, 33]. We observed that removing potential outliers can improve IR. However, this process may also inadvertently reduce the number of inliers, which can negatively affect registration [57]. Therefore, we also report the Inlier Number (IN). For registration evaluation, following [9, 57], we employ mean Rotation Error (RE), and Translation Error (TE) of successful registered cloud pairs as the evaluation metrics. We use Registration Recall (RR) to denote the registration success rate, which is defined as the fraction of point cloud pairs with RE$\leq 5°$, TE $\leq 60cm$ on the KITTI, and RE$\leq 15°$, TE $\leq 30cm$ on the ScanNet [9, 45, 57].

4.2 Keypoint Matching on Outdoor

Our method is equipped with several widely-used matchers, including the NN, OT and Mutual Nearest Neighbor (MNN) [2, 23]. Recent studies [9, 57] had shown that FCGF [12] exhibits little difference compared with FPFH [35] on the KITTI dataset. Therefore, we choose FPFH and GEDI [31] as the local geometric descriptors for evaluation.

The correspondence evaluation results are shown in Tab. 1. Clearly, our method significantly enhances all baseline matchers. For example, when applying the FPFH and GEDI in the easy dataset, the IN of NN matcher is 5 and 2 times improved after being equipped with our ML-SemReg. The reasons are as follows: (1) The Group Matching with Local-SS enhances the matcher's capability to perceive local-level semantic information, which can prevent inconsistent correspondences. (2) The issue of missing matches between classes is mitigated by Group Matching, as it allows a single keypoint to be matched with multiple

Table 1: Keypoint matching comparison on KITTI.

Matcher	FPFH						GEDI					
	easy		medium		hard		easy		medium		hard	
	IN	IR(%)	IN	IR(%)	IN	IR(%)	IN	IR(%)	IN	IR(%)	IN	IR(%)
MNN	81.37	8.81	20.85	2.50	6.64	0.81	457.70	34.16	156.46	13.45	51.55	4.72
NN	241.28	4.83	72.75	1.45	25.11	0.50	1024.21	20.48	376.06	7.52	129.33	2.59
OT	228.98	4.58	73.52	1.47	25.10	0.50	958.01	19.16	354.38	7.09	120.98	2.42
MNN+Ours	450.27	**38.10**	181.47	**18.41**	61.33	**7.56**	868.35	**60.41**	389.18	**34.33**	140.77	**14.10**
NN+Ours	**1246.10**	20.33	556.59	8.69	**231.35**	4.01	**2046.21**	33.99	1020.55	16.40	414.38	6.35
OT+Ours	1240.34	20.01	**580.45**	9.13	229.59	4.64	2031.18	33.13	**1040.10**	17.10	**444.18**	7.21

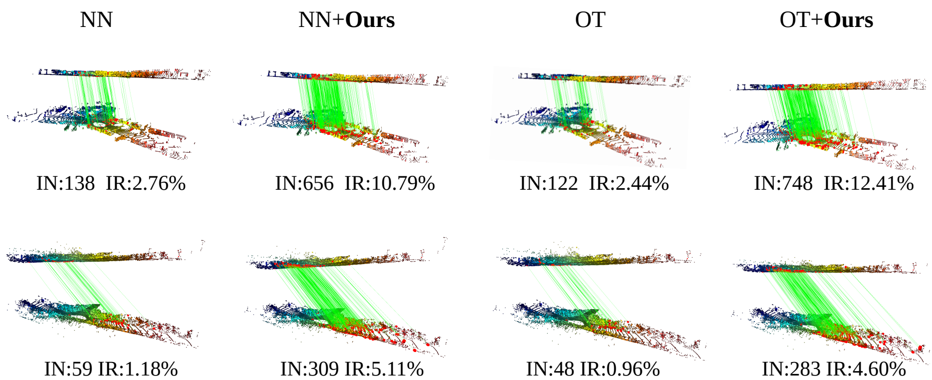


Fig. 4: Qualitative comparisons on the KITTI dataset of NN, NN+Ours, OT, and OT+Ours matchers using FPFH descriptor. The first and second rows depict examples from the easy (10-20m) and hard (20-30m) datasets, respectively. Inliers (residuals $< 0.5m$) are indicated by green lines. Best viewed on screen.

keypoints. (3) SS-Consistency is achieved by the MM module using the scene-aware descriptor BMR-SS. This reduces the matching candidate set to a refined subset, thus increasing the matching hit rate. In addition, the findings highlight that in both the medium and hard datasets, the IR of NN+Ours with FPFH is higher than that of NN with GEDI. This suggests that traditional descriptors equipped with our ML-SemReg can easily catch up or even surpass learning-based ones. We show qualitative results in Fig. 4. It is observed that the NN and OT matchers provide significantly more accurate inliers when equipped with ML-SemReg.

4.3 Pairwise Registration on Outdoor

In this section, we use correspondences created by NN and NN+Ours in Sec. 4.2 to estimate the rigid transformation. We employ RANSAC [17], TeaserPlus [47], SC2PCR [9], MAC [57], and PointDSC [45] as the registration estimators, in which we use RANSAC1M and RANSAC4M to represent the RANSAC with 1

Table 2: Registration results on the KITTI with FPFH and GEDI descriptors.

Descriptor	Estimator	easy			medium			hard		
		RE(°)	TE(cm)	RR(%)	RE(°)	TE(cm)	RR(%)	RE(°)	TE(cm)	RR(%)
FPFH	RANSAC1M [17]	1.71	33.17	19.13	1.79	37.48	1.25	-	-	0.00
	RANSAC4M [17]	1.73	33.20	42.60	1.56	38.82	3.38	1.15	50.36	0.20
	TeaserPlus [47]	0.83	15.93	83.92	1.47	29.34	32.83	1.39	36.36	2.85
	MAC [57]	0.68	14.64	88.60	1.48	32.26	35.50	1.89	31.46	1.43
	SC2PCR [9]	0.35	10.15	90.69	0.82	22.45	63.17	1.01	31.53	8.56
	PointDSC [45]	0.35	10.24	89.64	0.81	22.28	50.05	1.10	35.09	6.99
	RANSAC1M+Ours	1.94	31.06	57.85	2.10	36.18	26.23	2.26	37.99	6.93
	RANSAC4M+Ours	1.91	29.98	58.71	2.04	35.35	33.98	1.95	37.18	11.10
	TeaserPlus+Ours	0.39	10.01	95.42	0.87	17.91	85.49	1.34	27.49	55.99
	MAC+Ours	0.44	7.91	96.43	0.69	14.70	87.01	0.95	25.71	56.33
	SC2PCR+Ours	0.29	7.00	96.49	0.51	12.61	**91.01**	0.70	21.01	67.82
	PointDSC+Ours	**0.24**	**6.81**	**96.61**	**0.44**	**12.31**	90.11	**0.69**	**19.99**	**68.13**
GEDI	RANSAC1M [17]	1.40	24.91	72.04	1.58	29.78	27.75	1.07	36.22	1.43
	RANSAC4M [17]	1.40	25.34	76.29	1.55	29.56	44.75	1.61	30.89	4.89
	TeaserPlus [47]	0.28	7.74	95.18	0.59	15.51	88.58	0.91	26.54	51.71
	MAC [57]	0.24	6.39	95.94	0.49	13.05	89.50	0.93	27.51	40.53
	SC2PCR [9]	**0.20**	4.80	96.32	0.36	10.74	89.60	0.55	20.94	55.86
	PointDSC [45]	**0.20**	**4.77**	96.18	0.37	10.79	91.69	0.53	20.50	50.50
	RANSAC1M+Ours	1.61	25.91	79.55	1.68	30.84	50.36	1.54	34.61	17.01
	RANSAC4M+Ours	1.60	26.30	77.59	1.91	31.88	59.48	1.79	36.21	24.34
	TeaserPlus+Ours	0.25	7.19	97.50	0.56	12.10	94.35	0.81	23.01	78.85
	MAC+Ours	0.22	5.79	97.62	0.46	11.00	94.99	0.60	20.99	74.65
	SC2PCR+Ours	**0.20**	5.20	**98.10**	**0.23**	**9.71**	96.23	0.50	17.81	81.20
	PointDSC+Ours	**0.20**	5.20	97.91	0.33	10.01	**96.31**	**0.44**	**16.43**	**82.01**

million and 4 million iterations, respectively. The registration results are reported in Tab. 2 for FPFH and GEDI, respectively.

Enhancing Registration Recall. From Tab. 2, we can draw the following conclusions: (1) Our method can boost traditional RANSAC-type estimators by a large margin since it can effectively improve the IR of correspondences. The number of iterations required for RANSAC to correctly estimate the model increases exponentially as the IR decreases. In practical applications, the maximum number of iterations of RANSAC is usually limited to save time, which leads to its frequent failure when the IR is low. (2) Our method can also improve the performance of SOTA estimators. The more difficult the registration, the more obvious the improvement. This is due to the fact that our method can simultaneously improve the IR and IN of initial correspondence. On the one hand, a higher IR ensures better algorithm convergence. On the other hand, a higher number of inliers improves the ability to identify outliers [48, 57], thereby enhancing the quality of correspondences. For example, our approach enhances the SC2PCR and MAC by 59 and 54 percentage points, respectively, when utilizing the FPFH descriptor on the hard dataset. (3) Our method shows good robustness in low-overlap scenarios, e.g., when employing SC2PCR, the RR of SC2PCR+Ours on the medium dataset reaches a remarkable 91.01%, even surpassing the RR of SC2PCR on the easy dataset.

Table 3: Results of correspondences and registration on the ScanNet.

Registrator	40% ∼ 50%					50%∼60%				
	IN	IR(%)	RE(°)	TE(cm)	RR(%)	IN	IR(%)	RE(°)	TE(cm)	RR(%)
RANSAC1M [17]	204.99	6.95	5.93	14.48	54.04	269.85	9.31	5.58	14.14	70.51
RANSAC4M [17]			6.02	14.86	59.58			5.68	14.20	71.33
TeaserPlus [47]			3.00	8.76	73.02			2.58	7.52	84.39
PointDSC [45]			3.13	9.78	73.11			3.18	8.01	76.31
MAC [57]			5.28	13.92	46.57			4.93	12.57	61.43
SC2PCR [9]			2.82	8.34	79.88			2.36	7.00	87.55
RANSAC1M+Ours	**326.32**	**9.32**	6.31	15.71	64.31	**443.32**	**12.03**	6.01	15.61	74.88
RANSAC4M+Ours			5.99	14.98	62.97			5.75	14.49	68.64
TeaserPlus+Ours			3.84	8.31	82.33			3.53	6.99	90.01
PointDSC+Ours			3.15	9.01	86.91			3.13	7.43	89.24
MAC+Ours			4.65	14.92	61.15			5.21	13.88	74.01
SC2PCR+Ours			**2.65**	**8.12**	**89.04**			**2.12**	**6.73**	**92.78**

Enhancing Handcrafted to Suppress Learning-based. Even without using any data tranining techniques, our method significantly improves the performance of FPFH (handcrafted) and is comparable to learning-based descriptors. As shown in Tabs. 1 and 2, e.g., using NN+Ours+SC2PCR with FPFH setting yields IR=8.69% and RR=91.01%, slightly outperforming NN+SC2PCR with GEDI descriptor in the medium case, where IR=7.52% and RR=89.60%.

Enhancing Registration Accuracy. As shown in Tab. 2, our algorithm shows higher registration accuracy in most scenarios. The reasons may be twofold: (1) ML-SemReg can generate correspondences with a higher IR, which is important for the convergence of registration estimators such as the MAC [57]. (2) Our method enhances the creation of distant correspondences, which is more conducive to minimizing the transformation error. Please refer to the visualization results in the supplementary material for details. We can also see that although the TE of SC2PCR and PointDSC using GEDI descriptor on the easy dataset slightly increases after using ML-SemReg, our RR is better. This may be due to the fact that the quality of correspondences is already good enough in this easy case, causing our method to not improve the registration accuracy.

4.4 Pairwise Registration On Indoor

The results of NN and NN+Ours with FPFH descriptor on the indoor ScanNet dataset are reported in Tab. 3. Again, our pipelines significantly outperform the original methods on all evaluation metrics. For instance, our method enhances the RR of SC2PCR by 9.16% for overlap ratios ranging from 40% to 50%, and by 5.23% for overlap ratios ranging from 50% to 60%.

4.5 Robustness to Semenatic Segmentation Models

To evaluate the robustness of ML-SemReg to semantic labels, we equipped it with five different semantic segmentation models (from 2019 to 2022). The correspondence evaluation results are shown in Tab. 4. We can clearly see that our

Table 4: Correspondence evaluation results when using semantic labels from different semantic segmentation models on the three types of datasets. Our method demonstrates stability across different models.

Method	mIoU	easy		medium		hard	
		IN	IR(%)	IN	IR(%)	IN	IR(%)
RangeNet++(2019) [29]	52.2	1877.45	29.81	873.51	13.44	321.11	5.31
KPConv(2019) [39]	58.8	**2056.72**	32.85	**1075.59**	16.44	425.34	6.61
Cylinder3D(2021) [60]	63.7	<u>2054.04</u>	32.53	<u>1052.47</u>	<u>16.54</u>	448.60	**6.88**
SPVNAS(2020) [38]	66.4	2034.78	<u>33.01</u>	1028.19	16.19	434.65	6.40
PTv2(2022) [43]	70.3	2029.30	32.42	1026.37	**16.55**	**451.40**	6.71
2DPASS(2022) [46]	72.9	2025.08	**33.03**	1037.57	16.37	<u>434.78</u>	<u>6.74</u>

method is stable across different models. This is because (1) the Local-SS employed by Group Matching exhibits robustness to segmentation noise, and (2) the landmark categories (such as car, pole, trunk) required for BMR-SS construction are easy-to-segment, thus the differences between different models are not significant. We also provide more analyses experiments in supplementary material.

4.6 Parameter Analysis Experiments

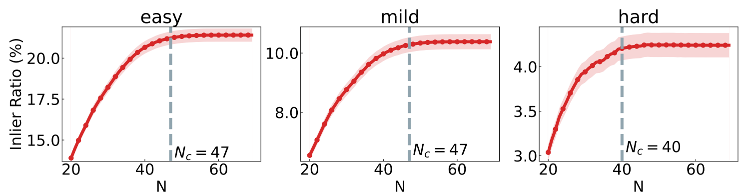
For better understanding of our ML-SemReg, we conduct comprehensive experiments to study the parameter sensitivities, including N and L of the BMR-SS, and K of the binary selection function $\mathrm{TL}^K(\cdot)$. We employ the NN+Ours with GEDI descriptor on the three types of KITTI datasets for study.

Receptive Field and Precision of BMR-SS. Fig. 5 illustrates our exploration of two key parameters of BMR-SS: (1) The ring number, denoted as N; (2) The ring length, denoted as L (m). N_c is defined as the first N value at which IR exceeds 99% of the maximum IR in Fig. 5a and L_t corresponds to the maximum IR in Fig. 5b. As depicted in Fig. 5a, IR gradually improves with the increase of N as the expansion of perceptual field, since more relevant scene information can be encoded in BMR-SS. We notice that when $N > N_c$, the perceptual field already encompasses almost all relevant semantic landmarks and thus only brings marginal improvements. N_c of hard dataset is lower than other two datasets. This may caused by oversized receptive fields which contain some uncorrelated landmarks in non-overlapping parts, thus negatively affecting BMR-SS. Moving to Fig. 5b, IR initially improves with the increase of receptive field. However, IR starts to decrease after surpassing L_t. This decrease may caused by large values of L, which result in a large number of landmarks within a ring that satisfy the SS-Consistency, thereby reducing localization accuracy. Despite the noticeable changes in IR with the variations of N and L, the optimal parameter settings remain stable, indicating the robustness of our method across diverse overlap scenarios.

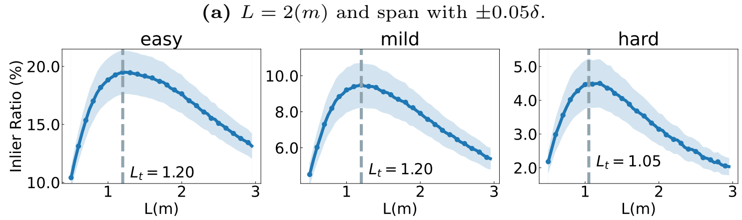
Balance of BMR-SS and Descriptor. The parameter K in the function $\mathrm{TL}^K(\cdot)$ is used to retain the top-K maximal values in each row of the scene

Table 5: Analysis the effect of K in TL^K function. Bold indicates the maximum value, underline indicates the second maximum, and '-' indicates not using ML-SemReg.

	K	-	1	2	3	4	5	6	7	8	9	10
FPFH	IN	172.01	129.17	**821.11**	<u>666.33</u>	489.48	323.80	289.17	233.82	200.29	186.12	167.21
	IR(%)	2.49	3.30	**13.10**	<u>10.77</u>	7.85	5.99	4.18	3.56	3.26	2.98	2.33
GEDI	IN	627.88	557.73	<u>1085.01</u>	**1151.11**	1025.11	882.98	791.44	741.32	682.88	665.33	636.12
	IR(%)	11.01	12.44	<u>17.98</u>	**18.68**	16.68	15.11	13.33	12.11	11.44	10.88	10.38



(a) $L = 2(m)$ and span with $\pm 0.05\delta$.



(b) $N = 30$ and span with $\pm 0.2\delta$.

Fig. 5: Sensitivity of IR to N and L. Each scatter is the mean of metrics, and δ is the standard deviation of all samples.

similarity matrix. Tab. 5 studies the impact of parameter K on matching performance using NN with FPFH/GEDI descriptors. We start with $K = 1$ and gradually increase it to 10. When $K = 1$, MM over-relies on the BMR-SS description, so our method performs even worse than the original matcher (NN). Our method achieves the best performance when $K = 2$ or 3. The results also show that any value of $K > 1$ outperforms the raw matchers. As K increases, the influence of the MM diminishes, causing the matcher to rely more and more on unstable local geometric features and leading to degradation. This suggests that local geometric features and BMR-SS can be balanced by K.

4.7 Ablation Study

We systematically evaluate the effectiveness of all components utilizing NN matcher, SC2PCR estimator, and FPFH/GEDI descriptors. We progressively integrate the GM and MM modules. Furthermore, we perform a comparative analysis between Matching in the Same Category (MSC) and matching in the matching group based on the LS-Consistency (i.e., GM). The results are shown in Tab. 6, from which we can draw the following conclusions: (1) Performance

Table 6: Ablation study on KITTI when leveraging NN matcher, SC2PCR estimator, and FPFH/GEDI descriptors. **MSC:** Match in the Same Category. **GM:** Group Matching. **MM:** Mask Matching.

		MSC	GM	MM	IN	IR(%)	RE(°)	TE(cm)	RR(%)
FPFH	0)				115.36	2.36	0.71	21.44	54.33
	1)	✓			74.30	1.51	0.81	25.00	52.56
	2)		✓		161.74	2.60	0.63	20.31	60.90
	3)			✓	<u>550.20</u>	**11.01**	<u>0.52</u>	<u>13.55</u>	<u>83.44</u>
	4)		✓	✓	**681.01**	<u>10.12</u>	**0.44**	**13.33**	**85.01**
GEDI	0)				508.41	11.01	0.40	12.41	83.77
	1)	✓			287.82	5.66	0.53	13.33	82.00
	2)		✓		615.73	9.89	<u>0.34</u>	12.25	83.99
	3)			✓	<u>1014.82</u>	**20.50**	<u>0.34</u>	<u>11.56</u>	<u>91.33</u>
	4)		✓	✓	**1143.41**	<u>18.80</u>	**0.32**	**11.31**	**91.88**

drops after using the MSC (comparing row 1 with 0), since the estimated semantic labels are inaccurate (semantic noise), producing new outliers and causing inlier missing. (2) The LS-Consistency overcoms the semantic noise problem and the GM module reduces the search space of correspondence establishment (comparing row 2 with 1 and 0), thus improving the IN and RR. (3) The MM module can significantly improve matching performance (comparing row 3 with 0), i.e., the RR is improved from 54.33% to 83.44% when employing the FPFH descriptor, and improved from 83.77% to 91.33% when employing GEDI. This improvement stems from its good scene awareness ability, which is complementary to local geometric similarity. (4) The combination of GM and MM modules achieves the best performance since they ensure multi-level semantic consistency, allowing them to overcome both inter-class and intra-class mismatching problems, thereby further improving the overall accuracy.

5 Conclusion

In this work, we presented ML-SemReg, a plug-and-play framework that is designed to enhance correspondence-based PCR by leveraging multi-level semantic consistency. To address inter-class mismatching, we introduced a novel Group Matching module, which restricts matching to groups that inherently satisfy Local Semantic Consistency. To suppress intra-class mismatching, we proposed a Mask Matching module based on our scene-aware Binary Multi-Ring Semantic Signature descriptor. Extensive experiments on indoor and outdoor datasets demonstrated that our method significantly outperforms state-of-the-art approaches and exhibits robustness to semantic labels.

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